

PROBLEM STATEMENT

CampusHire – Smart Campus Internship & Placement Management Platform

1. Title

CampusHire – Smart Campus Internship & Placement Management Platform

2. Domain

HRTech / Education Technology

3. Who is the user?

User Role	Purpose
Student	Creates a profile, searches internships, applies for opportunities, and tracks applications.
Company	Creates a company profile, posts internship opportunities, and manages student applications.
Placement Admin	Monitors students, companies, internships, applications, and placement activities.

4. What problem are we solving?

Students often depend on multiple channels to discover internship and placement opportunities, understand eligibility, submit applications, and track progress. Companies and placement administrators also need a structured way to publish opportunities and manage applications. This can cause scattered information, manual tracking, delayed communication, and difficulty monitoring recruitment activities. CampusHire addresses this by providing a centralized platform for students, companies, and placement administrators.

5. Proposed Solution

CampusHire is a full-stack web application that centralizes campus internship and placement activities. Students can maintain profiles, browse opportunities, and submit applications. Companies can publish internship opportunities and manage applications. Placement administrators can monitor users and platform activities. The application uses React.js, Java 17 Spring Boot, REST APIs, Spring Security with JWT, Spring Data JPA/Hibernate, and MySQL.

6. Core Entities / Database Tables

Entity / Table	Purpose
User	Authentication and common account information.
Student	Student-specific profile information.
Company	Company and recruiter information.
Internship	Internship opportunity and eligibility details.
Application	Student applications for internship opportunities.
Interview	Interview schedules linked to applications.
Offer	Selection and offer information for successful candidates.

7. User Roles & Permissions

- **Student:** Register/login, manage profile, view internships, apply, and track applications.
- **Company:** Register/login, manage company profile, create internship opportunities, and view/manage applications.
- **Placement Admin:** Monitor users and opportunities, manage platform records, and view placement activities.

8. Success Criteria

- Secure registration and login using JWT-based authentication.
- Students can view internships and submit applications.
- Companies can create and manage internship opportunities.
- Data is persisted in MySQL through Spring Data JPA/Hibernate.
- React communicates with Spring Boot through REST APIs.
- Role-based access controls protected functionality.
- The design supports future interview, offer, notification, and AI recommendation features.

9. Out of Scope

- Direct integration with a college's official placement management system.
- Real-time integration with external company HR systems.
- Guaranteeing an internship or job placement.
- Advanced AI recommendation functionality in the initial MVP.
- Production-scale analytics and advanced deployment features in Review-I.

10. Chosen Track

Java Track

Backend: Java 17 + Spring Boot 3.x

Security: Spring Security + JWT

ORM/Data: Spring Data JPA + Hibernate

Database: MySQL 8

Build: Maven

Testing: JUnit 5

API Documentation: Springdoc OpenAPI / Swagger UI

Frontend: React.js + Axios + Bootstrap/Tailwind CSS

Review-I MVP Focus: JWT authentication and two end-to-end flows: company creates an internship; student views and applies for an internship, with data flowing React → Spring Boot REST API → MySQL → React.